Reward Shaping Is Noise Shaping

and a model of the accuracy–diversity frontier of RLVR

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The question

- RL with verifiable rewards: sample answers, a checker grades them, update $g = \sum_i A_i \nabla \log \pi(y_i)$.
- Many papers modify the advantage A_i (PSR: $A = r$; NSR: $A = r - 1$; constant baselines; pass@k weights; entropy terms) and report higher **pass@k**.
- They compare methods **at equal training steps**, which confounds pass@k with progress in pass@1.
- We compare **pass@16 at matched pass@1**: linear interpolation along Dr. GRPO's trajectory and temperature envelope, bootstrap over problems.

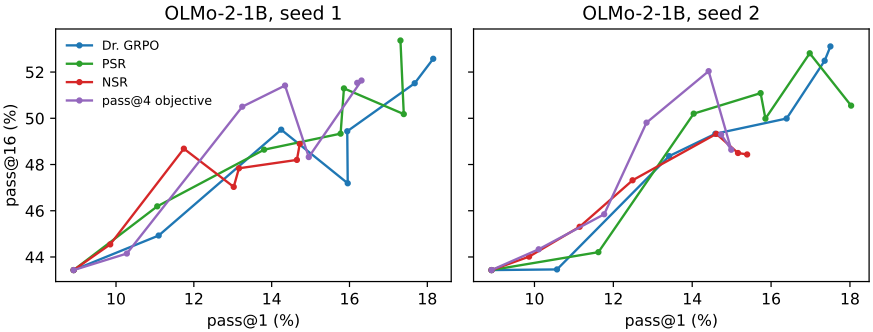
Theory: per-problem direction, rule-dependent noise

For any correctness-only advantage rule, per problem x with $p_x = \Pr_\pi(r = 1 \mid x)$:

$$\mathbb{E}g_x = c(p_x)\,\nabla p_x, \qquad \mathbb{E}\|g_x\|^2 = F_1 D_1(p_x) + F_0 D_0(p_x) + \|\nabla p_x\|^2 X(p_x)$$

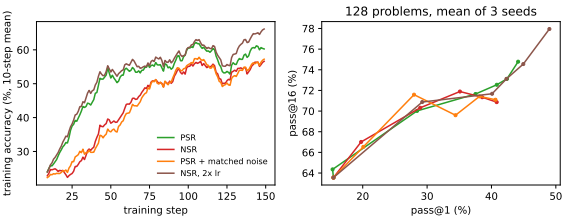
- A rule changes only the per-problem coefficient and the second moment; F_0/F_1: 2.14–2.54 (OLMo), 2.7–3.4 at initialization, 1.4–1.6 later (Qwen).
- Cross-problem reweighting can rotate the aggregate update (shared parameters), so a common curve is an empirical question.
- At matched pass@1, $P_k = N^{-1}\sum_x [1 - (1 - p_x)^k]$ depends only on the dispersion of $\{p_x\}$.

Diagnosis 1: one curve, two model families



OLMo: NSR minus PSR at matched pass@1 -0.30 pp, 90% $[-1.28, +0.77]$: average equivalence within ± 2 under the tested protocol. On Qwen, PSR, NSR, five baselines, the NSR paper's implementation and a pass@4 objective lie within about ± 2 pp.

Diagnosis 2: the noise is the cause



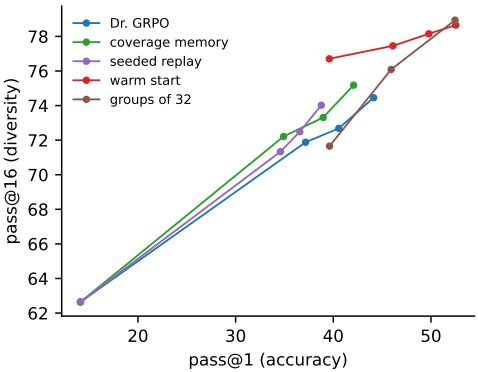
- Add trace-matched zero-mean sign noise to PSR.
- It reproduces NSR's pooled early pass@1 lag: $\phi = -0.11$ $[-0.22, -0.05]$ (NSR $= 0$, PSR $= 1$); one seed heterogeneous.
- NSR at $2\times$ learning rate reaches PSR's original rate.
- Temperature control: a late checkpoint sampled hotter does not exceed the trajectory (-1.42 pp).

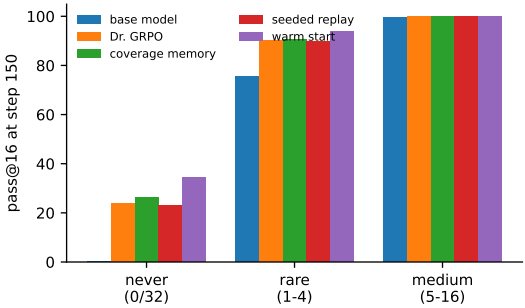
Interventions intended to move the frontier

Intervention	gain at matched pass@1 (pp of pass@16)	
Reward shaping, entropy, unlikeliness, pass@k objective	≈ 0	finished
Replay of own successes (coverage-weighted)	$+1.19$ $[+0.54, +1.79]$	finished
Replay seeded with base-model successes	$+1.02$; step-150 pass@1 -5.35	finished
Annealed replay; MaxRL + completion (pilot, 1 seed)	$+1.30$; $+1.12$; pass@1 -6.6, -6.5	interim
Group completion; prefix-guided rollouts (pilot)	–	evaluating
Warm start (supervised on base successes, then RL)	undefined: outside Dr. GRPO's range	
Groups of 32 ($4\times$ rollouts)	undefined: outside Dr. GRPO's range	2 of 3 seeds

Inside RL all gains are at most about $+1.2$ pp: below the registered $+2$ pp bar (pilot arms also fail its pass@1 allowance).

Trajectories and difficulty strata





Warm start and groups of 32 end near pass@1 52.5 (Dr. GRPO 44.1). The warm start's pass@16 increase is concentrated in the $c = 0$ stratum (24.0 $\rightarrow$ 34.2). At 1,024 tokens Dr. GRPO's step-150 pass@1 is 57.7 (vs 44.6 at 512): absolute values depend on the budget.

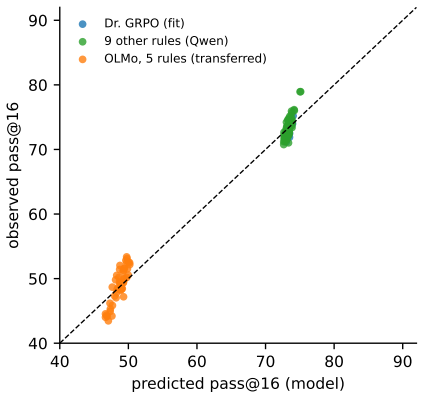
A model of the frontier

Per-question log-odds learning from base counts

$$\theta_x(0) = \log \tfrac{c_x+a}{n-c_x+b}, \qquad \dot{\theta}_x = p_x(1-p_x) + \tau\,\overline{p(1-p)}$$

- Inputs: the base model's per-question success counts; each checkpoint's pass@1 (sets progress t).
- $\tau = -0.20$ fitted once on Dr. GRPO, then **frozen** (pre-registered).
- Predicts P_k for $k > 1$, stratum-level P_{16} and per-problem count likelihoods.
- State-dependent log-odds dynamics: under a uniform log-odds shift the $c = 0$ stratum would reach 70–80 pass@16; observed 24.

Pre-registered frozen-model tests



Test set (checkpoints)	RMSE pass@16 (pp)	criterion
9 other rules, Qwen (56)	**1.21**	met
OLMo, τ from Qwen (50)	**1.90**	met
1,024 tokens (4)	**0.85**	met
Warm start (9)	3.56	not met (> 3)
450 steps (extrapolation)	pending	

Alternatives: uniform-shift RMSE 9.04–21.29, affine 1.27–12.58. Higher per-problem likelihood than affine in 117/119 checkpoints. OLMo stratum residuals: $+6.03$ ($c = 0$), -6.79 (rare).

Reading

- Within these settings, the matched-pass@1 frontier of policies trained from the base model is reproduced by the **base count distribution** and per-problem dynamics with weak transfer.
- In the model the frontier depends on two quantities: the **initial count distribution** and the **transfer coefficient** τ; advantage reweighting changes neither.
- **Warm start**: $c = 0$ stratum residual -11.92 pp; refitted $\tau = -0.65$ to -0.70 in all seeds: the supervised stage changes the subsequent dynamics.
- Against the model's base-initialized trajectory: warm start $+3.3$ to $+3.6$ pp, groups of 32 $+3.9$ pp at step 150, Dr. GRPO -1.2 to $+0.5$; T3 measures the extrapolation error.

Position in the 2026 literature

Not found in surveyed papers

- noise sets the progress rate along one curve
- NSR on PSR's curve, two families
- the noise-injection causal test
- matched-pass@1 protocol
- the temperature control
- a frozen-parameter frontier model

Prior papers and preprints

- same direction, per-question weight (Davis & Recht; Heckel et al.)
- base-seeded replay (TAR); coverage weight (BBG)
- rescue of all-wrong groups (LSPO, LUFFY)
- transfer curricula (TAC, GradAlign)
- "RLVR is a Kernel": main challenge to the frontier model

None of the surveyed method papers reports matched-pass@1 evaluation.

Where we stand

Stage	Independent review	Why
Diagnosis only	5/10	theory partly prior work
+ frontier model passes remaining tests	7/10 (scenario)	predictive result
+ method improving the frontier across scales	8/10 (scenario)	diagnosis → model → method

- **Running:** 450-step Dr. GRPO (extrapolation test); deep base sampling (sign and uncertainty of τ); TAR, BBG, MaxRL, W-REINFORCE at matched pass@1; pilot evaluations; 1,024-token and transfer checks.
- **Method candidate from the model:** allocate demonstration acquisition to problems the model predicts will stall; compare with random allocation and the warm start at equal compute.